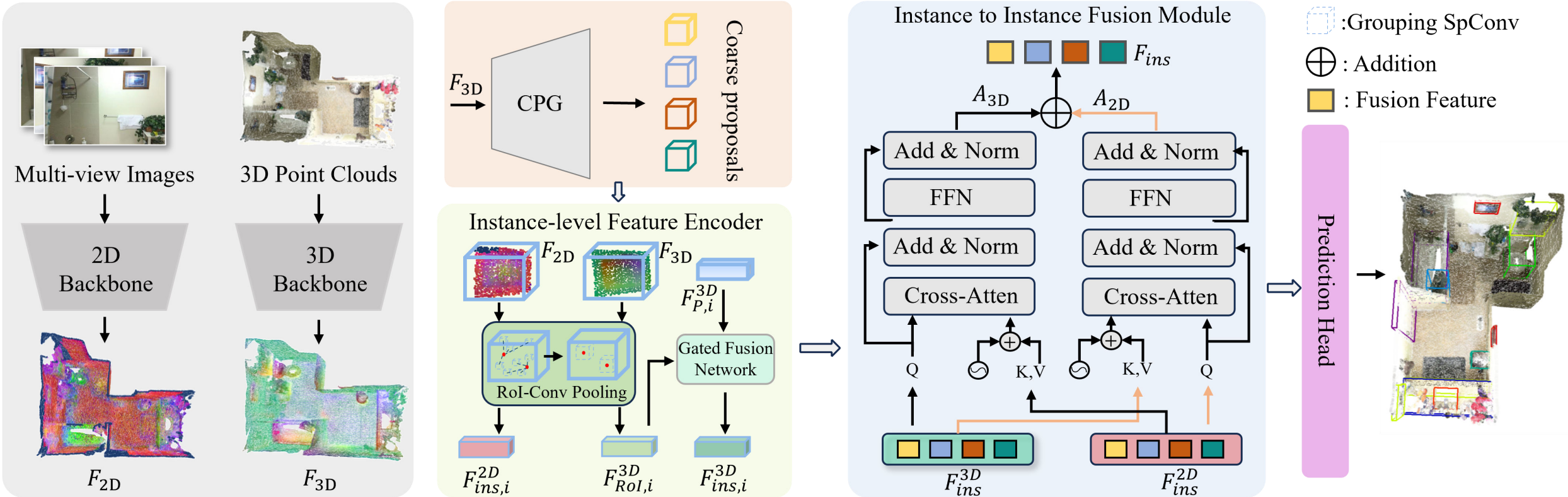
Multi-view Images
3D Point Clouds
2D Backbone
3D Backbone
F_{2D}
F_{3D}
F_{3D}
CPG
Coarse proposals
Instance-level Feature Encoder
F_{2D}
F_{3D}
$F_{P,i}^{3D}$
RoI-Conv Pooling
Gated Fusion Network
$F_{ins,i}^{2D}$
$F_{RoI,i}^{3D}$
$F_{ins,i}^{3D}$
Instance to Instance Fusion Module
F_{ins}
A_{3D}
A_{2D}
Add & Norm
FFN
Add & Norm
Cross-Atten
Q
K,V
K,V
Q
F_{ins}^{3D}
F_{ins}^{2D}
Prediction Head
:Grouping SpConv
: Addition
: Fusion Feature